# Provenance model methods and results outline

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## 1 Methods

1. We selected all the datasetIDstudy that had more than one provenance per species ($n = 5959$)
2. After Victor's decision rules and cleaning the dataset we end up with 31 single species from 23 studies.
3. We fit the provenance model with and without forcing first, compared model estimates and moved on with the model without forcing with chilling and time as the main predictors.
4. We performed posterior predictive checks for the provenance model without forcing.

## 2 Results

We fitted a provenance model for each species with more than one provenance.

1. Did forcing change the germination response?
   (a) We fitted the provenance model with forcing and without forcing.
   (b) Forcing does not seem to explain the variation between the no-forcing and forcing models. Neither the maximum temperature (Figure 1), the number of unique germination treatments (Figure 2) or the temperature interval seem to explain the observed variation (Figure 3) between the restricted vs full dataset.
   (c) The estimates were unchanged when fitting the no-forcing model with the most restricted data ($n = 1116$) and the full dataset ($n = 1219$).
   (d) We fitted the model on the most restricted dataset, with and without provenance, which did not change the estimates (Figure 4)
2. How did time, chilling and forcing impact germination? The below are for the model that includes forcing. We need to change the x axis.
   (a) Forcing had little effects on species, but increased germination for Maackia taiwanensis and Betula utilis and Picea glauca (Figure 6).
   (b) Chilling increased germination for most species with huge variations. Four species probability to germinate decreased in response to chilling (Figure 5).
   (c) Time had a similar trend to forcing, with some more species increasing germination in response to time. Most of them didn't shift (Figure 7).
   (d) We show posterior predictive checks averaged across all species and treatments with time (Figure)
3. Was there more variation across species or across provenances of the same species?
   (a) Germination varied more across species ($\sigma = 2.943$ (UI: 2.415, 3.552)) than across provenances ($\sigma = 0.556$ (UI: 0.45, 0.676))

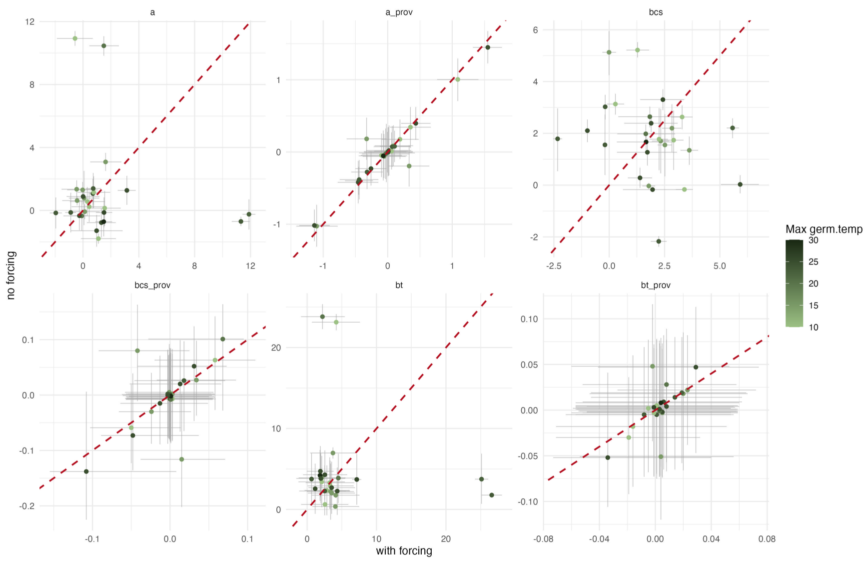


Figure 1: Datasets with forcing vs no forcing color coded by max temperature

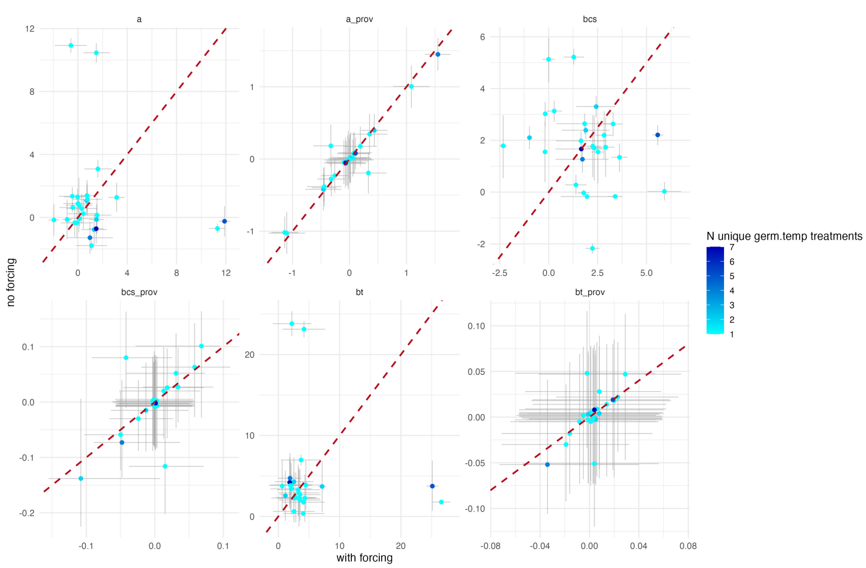


Figure 2: Datasets with forcing vs no forcing color coded by number of unique germ temp treatments

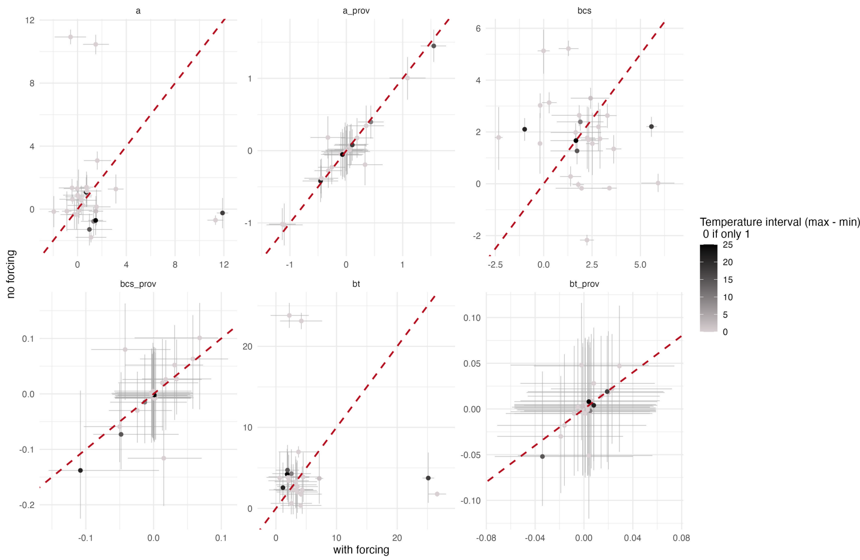


Figure 3: Datasets with forcing vs no forcing color coded by the temperature interval in the study

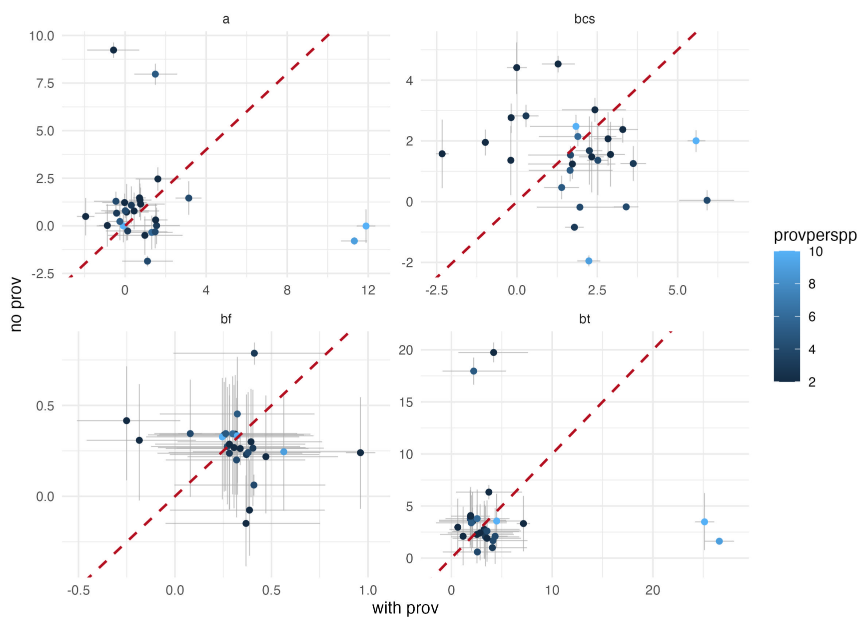


Figure 4: 1:1 line plot to compare the model that includes provenance and excludes provenance.

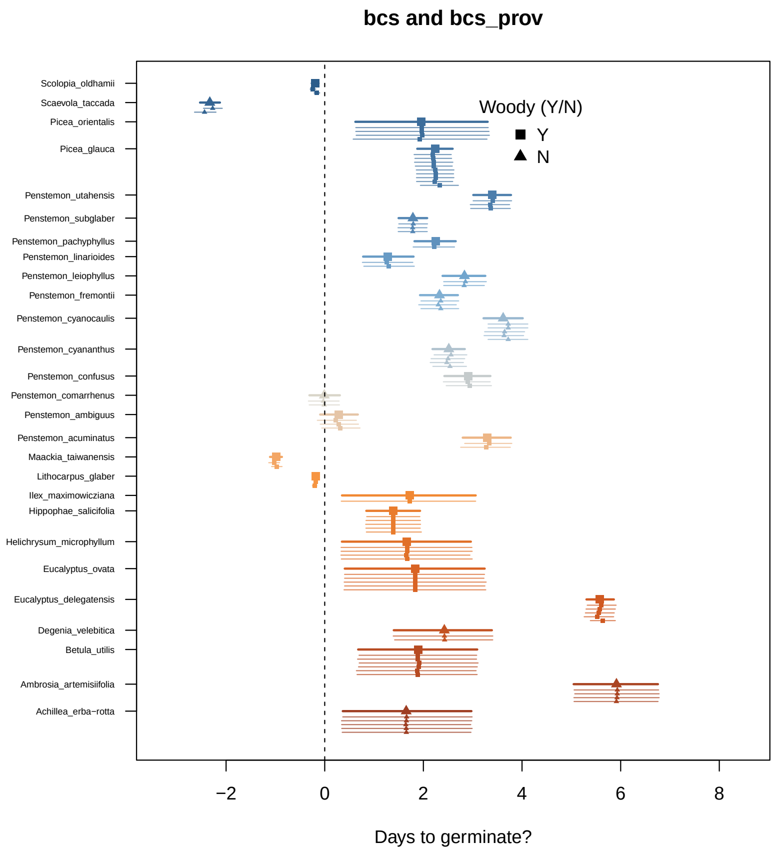


Figure 5: Chilling slope parameters

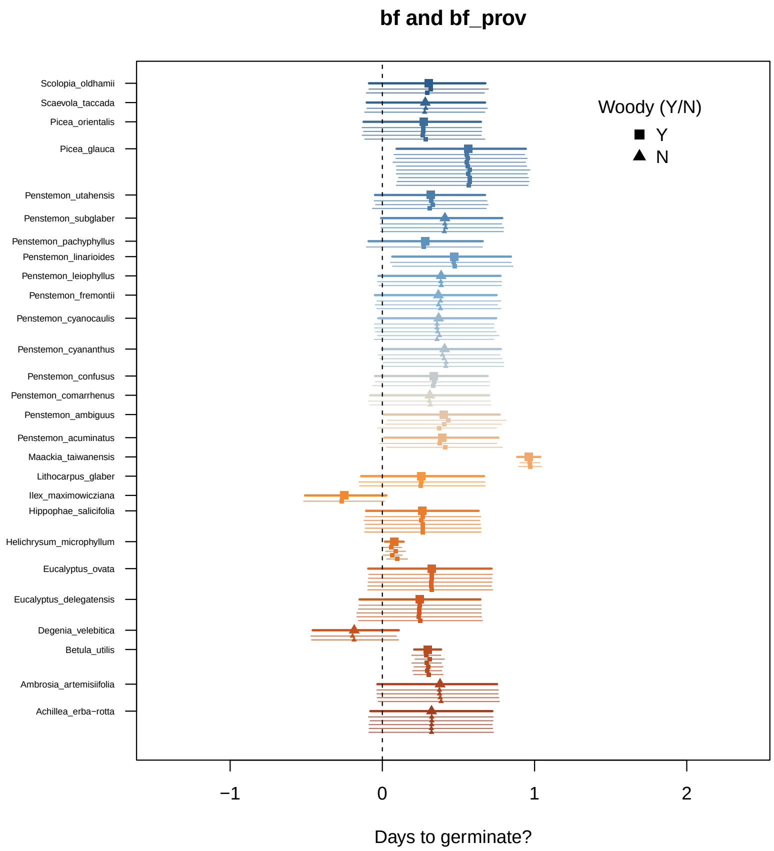


Figure 6: Forcing slope parameters

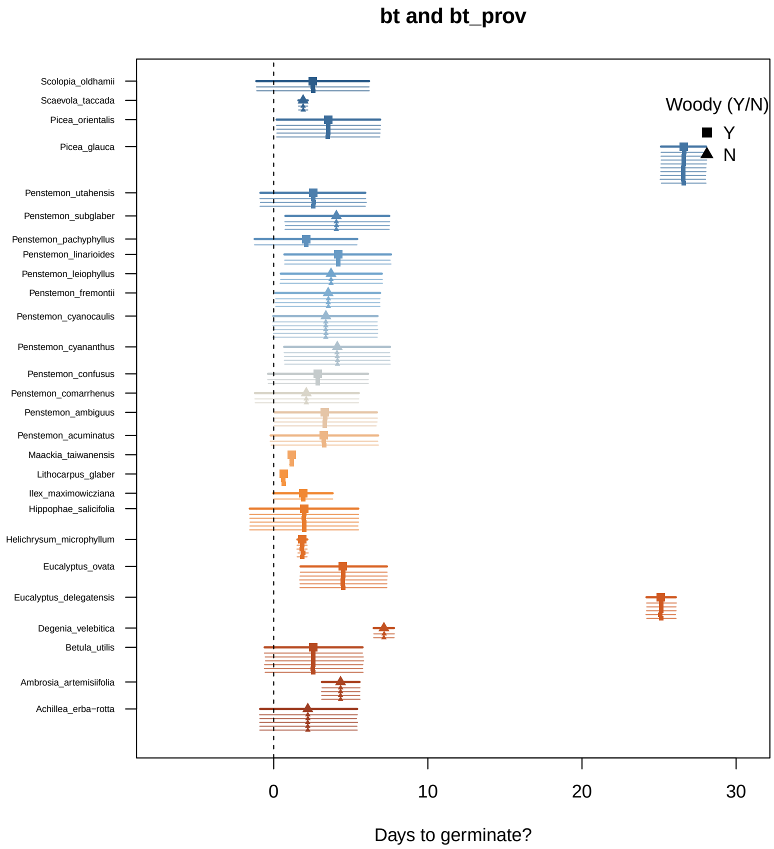


Figure 7: Time slope parameters

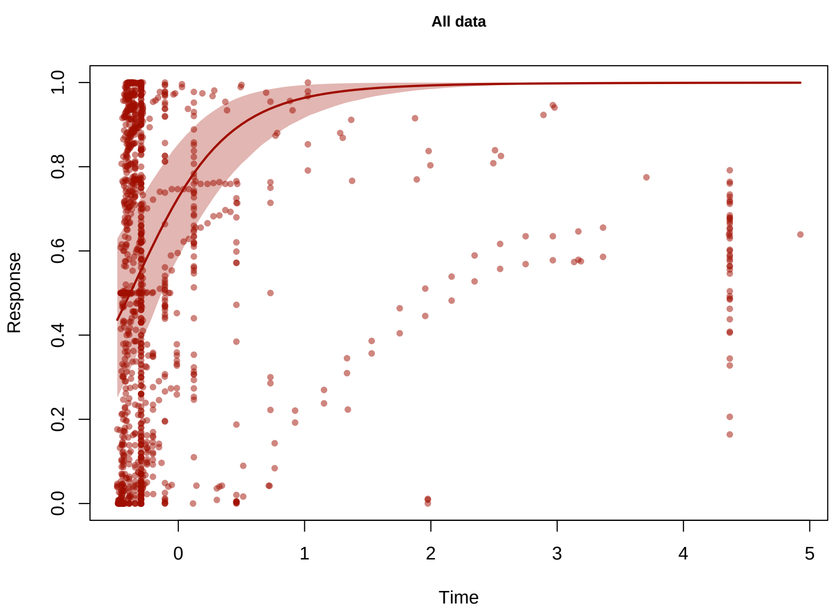


Figure 8: General response curve across all species as a function of chilling (scaled). We averaged the contribution of time and included it in the response curve.

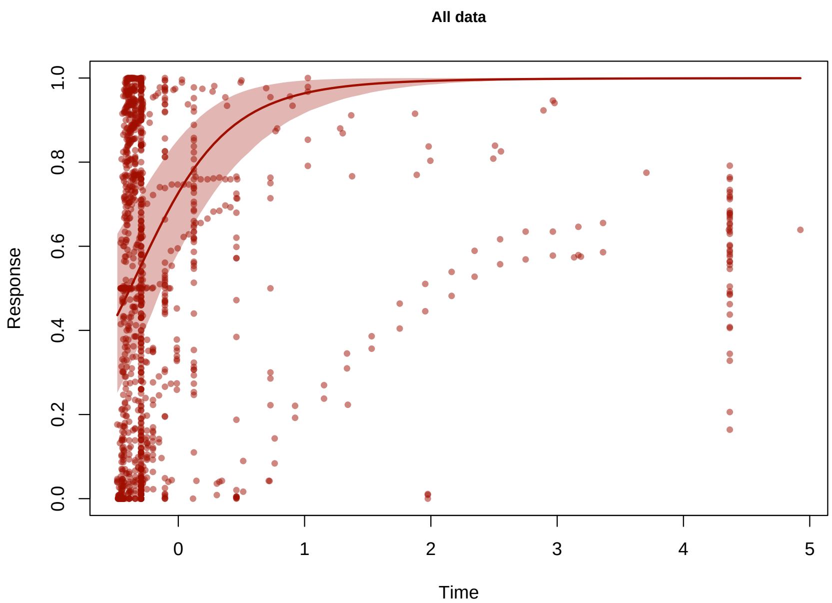


Figure 9: General response curve across all species as a function of chilling (scaled). We averaged the contribution of time and included it in the response curve.